Internship Worklog - Week 4

Name: Dhruv Jain (Data Intern)
Reporting Period: 02 November - 06 November 2026

Monday, 02 November 2026 - Access Policy Alignment & KT Coordination

- Transitioned focus from frontend module exploration to backend data architecture.
- Coordinated with Shamitr Sudhir Mardikar and Prudvi Pratap Kanakala to schedule the Knowledge Transfer (KT) session for the new database access system.
- **Compliance & Policy:** Thoroughly reviewed the revised strict database access policy. Acknowledged the mandatory 5-step workflow (CS Approval → ClickUp Request → PM Assignment → Temporary Access via Piyush Anand → Report Generation).
- **Heavy Task:** Drafted a personal Data Compliance Checklist to ensure all future query executions and data extractions strictly adhere to the new security protocols without causing operational delays.

Tuesday, 03 November 2026 - Knowledge Transfer & AWS Infrastructure Mapping

- Attended the comprehensive database KT session led by Shamitr Sudhir Mardikar and Gurdeep Singh.
- **Technical Focus:** Gained deep insights into the underlying AWS infrastructure (EC2/RDS) hosting the Digii Campus databases. Mapped the secure routing protocols and data extraction pipelines.
- Successfully negotiated secure access to download the foundational 10GB production-replica database file.
- **Heavy Task:** Analyzed the cloud database architecture to design a localized mirrored environment, ensuring that local query performance testing would accurately reflect production server constraints.

Wednesday, 04 November 2026 - 10GB Database Setup & Performance Tuning

- Focused on environment configuration to securely host and query the massive dataset locally.
- **Data Engineering:** Engineered the restoration process for the 10GB SQL dump. Overcame initial import bottlenecks by tuning SQL buffer pool sizes, max_allowed_packet configurations, and timeout parameters to handle the massive data volume.
- Conducted initial diagnostic queries and EXPLAIN plans to verify data integrity, evaluate index performance, and establish a baseline for query execution times.

Thursday, 05 November 2026 - Schema Audit & Data Anomaly Detection

- **Data Profiling:** Executed a deep dive into the schema, utilizing reverse-engineering tools to map the primary/foreign key relationships across core modules (CHC, Mess, Access Management).
- Authored complex multi-table SQL JOIN queries to stress-test the schema and validate the data workflows driving the frontend BI portal.
- **Enhancement Ideation:** Conducted a data quality audit. Identified potential historical data anomalies (e.g., orphaned transactional logs in Access Management) and proposed adding strict foreign key constraints and timestamp intervals to make the tables robust for predictive analytics.

Friday, 06 November 2026 - Managerial Review & Audit Presentation

- Participated in a strategic meeting with Mr. Suraj Bhatt to jointly explore the newly set up 10GB database and review the schema mappings.
- **Heavy Task:** Presented the findings from Thursday's Data Quality Audit, specifically highlighting the identified data anomalies and the proposed indexing strategies to optimize future report generation.
- Aligned on how the new strict database access policy will shape the workflow for future real-time report requests assigned via ClickUp.
- **Outcome:** Secured managerial alignment on the proposed schema enhancements and finalized the data analytics roadmap for the upcoming week.

Week 4 Summary

During the fourth week, the internship focus shifted significantly into rigorous Data Engineering and Database Administration. Beyond simply acquiring a 10GB database file, the core achievements included tuning local database configurations to handle massive data imports, auditing AWS infrastructure parameters, and enforcing enterprise-grade data compliance policies. By conducting a deep-dive schema audit, identifying data anomalies, and authoring complex stress-test queries, I successfully bridged the gap between frontend user workflows and the backend data architecture, laying a highly optimized foundation for advanced predictive analytics.